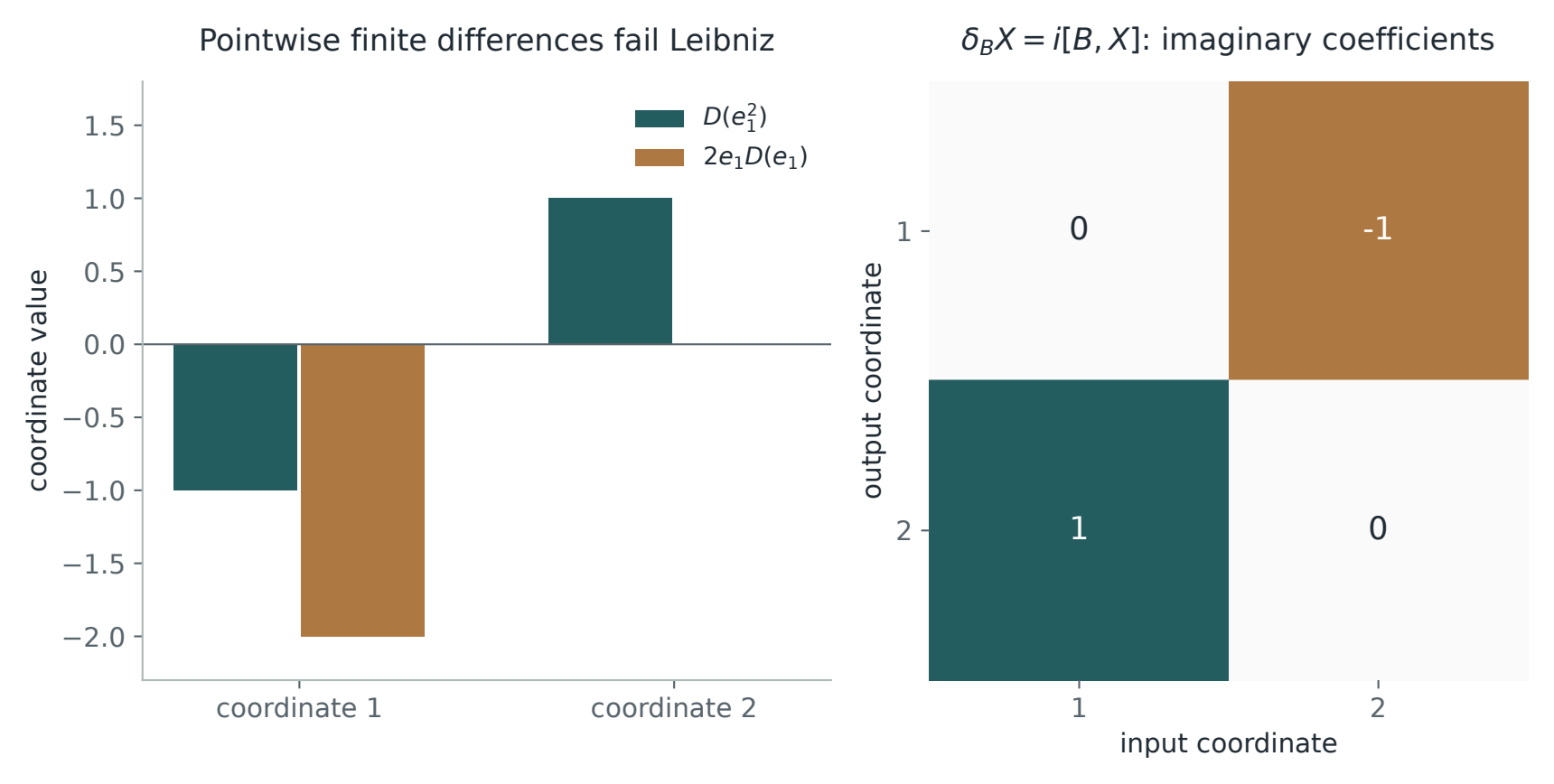
Pointwise finite differences fail Leibniz
$D(e_1^2)$
$2e_1D(e_1)$
coordinate value
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
coordinate 1
coordinate 2
$\delta_B X = i[B, X]$: imaginary coefficients
output coordinate
1
2
0
-1
1
0
1
2
input coordinate